

Jayamini Dissanayake

BSc (Hons) Computer Systems Engineering Undergraduate
Sri Lanka Institute of Information Technology (SLIIT)

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

Computer Systems Engineering undergraduate at Sri Lanka Institute of Information Technology (SLIIT) with hands-on experience in embedded systems, IoT solutions, software development, networking, and control systems. Skilled in Python, C/C++, Java, Raspberry Pi, Arduino, sensor integration, microcontrollers, and hardware–software interfacing, with practical experience in designing, developing, testing, debugging, and troubleshooting real-world engineering solutions. Strong foundation in system integration, automation, PCB design, Linux, network monitoring, and real-time control systems, demonstrated through academic and personal projects. Seeking an internship opportunity in Systems Engineering, IT Infrastructure, Networking, Embedded Systems, or Technology Project Management to apply technical knowledge, gain industry experience, and contribute to innovative technology solutions.

EDUCATION

BSc (Hons) in Computer Systems Engineering	
Sri Lanka Institute of Information Technology (SLIIT)	2024 – 2027
G.C.E. Advanced Level (A/L)	
Physical Science Stream	2022

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, SQL
Embedded Systems: Raspberry Pi, Arduino, AVR Concepts, Sensor Integration, Microcontrollers
Networking & Infrastructure: TCP/IP, SNMP, Network Monitoring, Network Troubleshooting, Linux, Zabbix
Web Development: Spring Boot, React, REST APIs, HTML, CSS, MySQL
Tools & Platforms: Git, GitHub, VS Code, Eclipse, Google Cloud Platform (GCP), EasyEDA
Concepts: Object-Oriented Programming (OOP), System Integration, Integrated Systems, Technical Documentation, PID Control, Automation, PCB Design, Debugging, Troubleshooting, System Testing

PROFESSIONAL EXPERIENCE

PID-Controlled DC Motor Stabilization System

April 2026

- Designed and implemented a PID-controlled DC motor speed stabilization system using Raspberry Pi and Python in a Linux environment.
- Developed closed-loop motor control with encoder-based real-time speed feedback and sensor processing.
- Tuned PID parameters (Kp, Ki, Kd) to achieve optimized system stability, low overshoot, and minimal steady-state error.
- Designed and integrated a custom PCB using EasyEDA for compact and reliable hardware implementation.
- Performed system modeling, gain tuning, disturbance testing, and performance evaluation for control system optimization.

Technologies: Python, Raspberry Pi, PID Control, EasyEDA

Line Following Arduino Robot Car

April 2026

- Designed and developed an autonomous line-following robot car using Arduino and embedded system principles.

- Implemented sensor-based path detection and real-time motor control for autonomous navigation.
- Developed embedded control logic using C/C++ and integrated IR sensors for accurate line tracking.
- Designed and tested hardware interfaces, motor driver control, and obstacle handling mechanisms.
- Applied embedded systems engineering concepts including sensor integration, control logic, and real-time system testing.

Technologies: Arduino, IR Sensors, Embedded C/C++

CradleBot – Smart Automated Baby Cradle System

April 2024

- Designed and developed an automated smart cradle system capable of detecting baby cries and initiating automatic rocking motion.
- Integrated sound and moisture sensors for infant monitoring and responsive system control.
- Implemented servo motor-based cradle movement and MP3 audio playback for soothing lullaby functionality.
- Developed embedded control logic using Arduino for real-time sensor processing and automation.
- Applied embedded systems, sensor interfacing, and automation principles to improve infant comfort and parental convenience.

Technologies: Arduino, Servo Motors, Sensors

AM Broadcast Receiver

October 2025

- Designed, built, and tested the audio output circuit of a basic AM broadcast receiver using discrete electronic components.
- Implemented multi-stage analog circuits including band-pass filtering and BJT-based signal amplification for audio recovery.
- Designed a common-emitter BJT amplifier with stable biasing to amplify weak detector signals for audio output.
- Performed circuit analysis, component selection, waveform testing, and signal observation using oscilloscope measurements.
- Constructed and tested the complete hardware circuit on a dot board with reliable wiring and circuit integration

Technologies: Analog Electronics, BJT Amplifiers, Filters, Oscilloscope Testing, Circuit Design, Dot Board Implementation

CERTIFICATIONS

Network Support & Security – Cisco Networking Academy (2026)

AI/ML Engineer – Stage 1&2 – SLIIT Faculty of Computing (2025)

Python Programming – University of Moratuwa, CODL (2025)

Web Design for Beginners – University of Moratuwa, CODL (2025)

Software Architecture – Udemy (2025)

Introduction to Software Engineering – Udemy (2025)

NMAP: Network Scanning & Recon (CNMP+) – OCSALY (2025)

SOFT SKILLS

Time Management

Communication Skills

Adaptability and Quick Learning

Teamwork and Collaboration

Problem-Solving and Analytical Thinking

Long term planning

References

Dr. Kapila D. Dissanayaka

Senior Lecturer (HG), Faculty of Computing

Sri Lanka Institute of Information Technology (SLIIT)

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Ms. Dinithi S. Pandithage

Lecturer | Year 2– CSNE Coordinator, Faculty of Computing

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